Roll No. ....

TEC–604

B. TECH. (ECE) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023

DATA COMMUNICATION NETWORKS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain OSI model with the help of a block diagram. (CO1)

OR

(b) What are the different network topologies available ? Explain and also state its advantages and disadvantages. (CO1)

2. (a) Explain basic concept of spread spectrum using block diagram. Differentiate between slow FHSS and fast FHSS. (CO1)

OR

(b) Write short notes on any *two* of the following : (CO1)

(i) Sky wave propagation

(ii) Packet switching

(iii) QAM

P. T. O.

3. (a) Explain the services provided by the data link layer. Discuss the different data link protocols based on the channel classification. (CO2)

OR

(b) What is Hamming bound ? A dataword of at least 11 bits is required. Find the values of k and n in the Hamming code $C(n, k)$ with $d_{min.} = 3$. (CO2)

4. (a) What is High-level Data Link Control (HDLC) protocol ? Explain its frame format in detail. (CO2)

OR

(b) What is piggybacking ? Explain Selective Repeat ARQ protocol. (CO2)

5. (a) Explain FDM. Ten signals, each requiring 4000 Hz, are multiplexed on to a single channel using FDM. How much minimum bandwidth is required for the multiplexed channel ? Assume that the guard bands are 400 Hz wide. (CO1, CO2)

OR

(b) Use the generator polynomial $g(x) = x^3 + x + 1$ to construct a systematic (7, 4) cyclic code. What are the error correcting capabilities of this code ? Construct the decoding table. If the received word is 1101100, determine the transmitted data word. (CO1, CO2)

Roll No. ....

TEC–603

B. TECH. (ECE) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

VLSI TECHNOLOGY AND DESIGN

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain silicon shaping. How can we identify the various types of Si wafers ? (CO1)

OR

(b) What do you understand by Epitaxy ? Explain Vapor Phase Epitaxy. Differentiate between Metallurgical Grade Silicon and Electron Grade Silicon. (CO1)

2. (a) What do you understand by Chemical Vapor Deposition ? (CO1)

OR

(b) Explain in brief about CZ Technology and their application. (CO1)

P. T. O.

3. (a) What do you understand by metallization ? Explain its application in VLSI Technology. (CO2)

OR

(b) Explain the annealing process and its application. (CO2)

4. (a) Explain the types of photoresists. (CO2)

OR

(b) Explain wet etching process in VLSI. (CO2)

5. (a) Explain in brief about reactive ion etching. (CO3)

OR

(b) What do you understand by CZ crystal growth technique ? (CO3)

Roll No. ....

TEC–602

B. TECH. (ECE) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, 2023

MICROWAVE ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Consider a length of Teflon-filled, copper K-band rectangular waveguide having dimensions a = 1.07 cm and b = 0.43. Find cut-off frequencies of the first three propagating modes. Draw the field patters for dominant TE mode. (CO1)

OR

(b) In a rectangular waveguide for which a = 1.5 cm, b = 0.8 cm, $\sigma = 0$, $\mu = \mu_0$ and $\varepsilon = 4\varepsilon_0$: (CO1)

$$H_x = 2\sin\left(\frac{\pi x}{a}\right)\cos\left(\frac{3\pi y}{b}\right)\sin(\pi\times 10^{11}t - \beta z)\,\mathrm{A/m}$$

For TE_{13} mode, determine :

(i) The cut-off frequency

(ii) The phase constant

(iii) The propagation constant

(iv) The intrinsic wave impedance.

P. T. O.

2. (a) What causes attenuation inside the waveguide ? Derive the expressions of the surface current components for TE_{10} mode in a rectangular waveguide having the following field components : (CO1)

$$H_z = A_{10} \cos\frac{\pi x}{a} e^{-j\beta z}$$

$$E_y = \frac{-j\omega\mu a}{\pi} A_{10} \sin\frac{\pi x}{a} e^{-j\beta z}$$

$$H_x = \frac{j\beta a}{\pi} A_{10} \sin\frac{\pi x}{a} e^{-j\beta z}$$

$$E_x = H_y = E_z = 0$$

OR

(b) If a resonant cavity is filled with a lossless material $(\mu_r = 1, \varepsilon_r = 3)$ and having dimensions a = 5.5 cm, b = 5.4 cm, and c = 5.1 cm is made of copper $(\sigma_c = 5.8 \times 10^7 \mathrm{S/m})$. Find the resonant frequency f_r and the quality factor for TE_{101} mode. (CO1)

3. (a) Explain the boundary conditions in a circular waveguide structure to obtain field equations in TE and TM mode. (CO1)

OR

(b) Find the cut-off frequencies of the first two propagating modes of a Teflon-filled circular waveguide with radius $a = 0.5$ cm. (CO1)

4. (a) Discuss the limitations of conventional tubes and need of microwave sources. (CO3)

OR

(b) Explain the working of two-cavity klystron and bunching of electrons process with proper diagram. (CO3)

5. (a) Discuss the basic structure of a Cylindrical Magnetron and how the electrons will follow cycloidal path in the cathode-anode space supported by suitable equations and diagrams. (CO3)

OR

(b) Discuss the difference between IMPATT and TRAPATT devices.(CO3)

Roll No. ....

TEC–601

B. TECH. (ECE) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023

WIRELESS COMMUNICATION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Derive an expression to show the relationship between co-channel reuse ratio (Q) and the cluster size. (CO1)

OR

(b) Discuss the trends of evolution from 1G to 4G that took place in Europe and us. (CO1)

2. (a) (i) The service area in a cellular system requires 12 repetitions of a cluster having 7 cells. Each cell has 16 channels. Find the total number of channels that can be simultaneously accessed in the system. (CO1)

(ii) A cluster has 4 cells each with 7 km^2 coverage area. The cellular system has to provide coverage over an area of 1765 km^2. Find the number of times the cluster has to be repeated.

OR

(b) (i) Explain meaning of the terms 'universal frequency reuse' and 'fractional frequency reuse'. (CO1)

(ii) Define dwell time and the factors on which the dwell time depends.

P. T. O.

3. (a) Assuming 7 cell cluster, find an expression to indicate SIR in co-channel cells in the downlink. (CO1)

OR

(b) A cellular system experiences path loss exponent to be 4, where a UE requires minimum 18 dB SIR to have acceptable voice quality. If there 7 cells in a cluster, will it support the minimum required SIR ? Now, assume the minimum required SNR is increased to 20 dB. Find the appropriate cluster size. (CO1)

4. (a) Why does sectorization enhance system capacity ? What do you mean by higher order sectorization ? (CO1)

OR

(b) Explain Hard, Soft and Softer hand-offs. (CO1)

5. (a) If a transmitter produces 50 watt of power uniformly in all direction, express the transmit power in units of (i) dBm, and (ii) dBW. If 50 watt is applied to unity gain antenna with a 900 MHz carrier frequency, find the received power in dBm at a free space distance of 100 m from the antenna. What is P_r (10 km) ? Assume unity gain for the receiving antenna. (CO2)

OR

(b) State significance of two-ray ground reflection model. Explain using suitable diagrams and obtain an expression for the received signal power. (CO2)

Roll No. ....

TEC–659

B. TECH. (ECE) (ESR & IoT) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023

ADVANCED EMBEDDED SYSTEM

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain the history of the Embedded System. Also, differentiate between General Purpose System and Embedded System. (CO1)

OR

(b) Explain the classification of Embedded System an discuss about the applications of it. (CO1)

2. (a) Explain the quality attributes of Embedded System. (CO1)

OR

(b) How can we design a currency counter using any microcontroller ? Explain in detail about the designing process of it. (CO1)

3. (a) Explain PIC microcontroller with the help of a block diagram. (CO1)

OR

(b) What is the role Timer and Counter in 8051 microcontroller ? Explain with a suitable example. (CO2)

P. T. O.

4. (a) Explain the role of the Program Status Word (PSW) in the 8051 microcontroller. (CO2)

OR

(b) Discuss the various addressing modes used in 8051 with the help of an example for each mode. (CO3)

5. (a) Write an 8051 assembly language program to toggle a port pin continuously. (CO3)

OR

(b) Discuss the interrupt system in the 8051 microcontroller. What is the difference between a maskable and non-maskable interrupt ? (CO3)

Roll No. ....

TOE–610

B. TECH. (ECE) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

OOPs USING C++

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain the features of Object Oriented Programming with examples.

(CO1, CO2)

OR

(b) Write a C++ program to create a class called StringDemo with the following members :

Data Members:

string sval

Function Members:

Constructor to initialize value of sval

void setvalue() → This member function will print value of sval

void printvalue() → This member function will extract and print all odd postion characters from the data member sval

P. T. O.

Example :

Sample input : Examination

Output : Eaiain

In main function create object of class StringDemo and perform the above task. (CO1, CO2)

2. (a) Write a C++ program to overload the post- and pre-increment operator (++). (CO2, CO3)

OR

(b) Illustrate different types of constructors in C++ with proper code. (CO2, CO3)

3. (a) What is the difference between static member function and instance member function ? Explain with the help of a C++ program. (CO1, CO2)

OR

(b) Explain function overloading in C++. Write a C++ program to demonstrate the concept of Compile time polymorphism. (CO1, CO2)

4. (a) With the help of C++ program, explain any four uses of scope resolution (::) operator. (CO1, CO3)

OR

(b) Define a class named SwapData with the following description :

Data Members

private int x, y

Constructor SwapData(int,int) → Constructor will store value in x and y.

Member function printval() → this member function will print value of x and y.

Friend function void swapval(SwapData &ob) → This friend function will swap value of x and y for the passing object. (CO1, CO3)

5. (a) Write a C++ program to overload binary + operator by using friend function. (CO2, CO3)

OR

(b) What is the output of the following program ? (CO2, CO3)

```
#include<iostream>
using namespace std;
class Exam
{
    int num;
    static int count;
public:
Exam()
{
    cout<<"10"<<endl;
    num= 10;
}
void input(int n)
{
    num = n + 10;
    count++;
}
void showValue()
{
    num++;
```

```
        cout<<num<<endl;
        cout<<count<<endl;
        cout<<Exam::num<<endl;
    }
};
int Exam::count;
int num=90;
int main()
{
    Exam obj;
    obj.input(5);
    Exam tob;
    tob.input(2);
    obj.showValue();
    tob.showValue();
    return 0;
}
```